

Estimating Active Dimension of Training Dynamics from One Scalar Log

with an Application to Grokking

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One log can reveal low-dimensional recurrent dynamics.

On the grokking runs studied here, it gives a transition signal; stored trajectories reveal a reversible change in covariance effective rank.

01 Dimensions and covariance effective rank

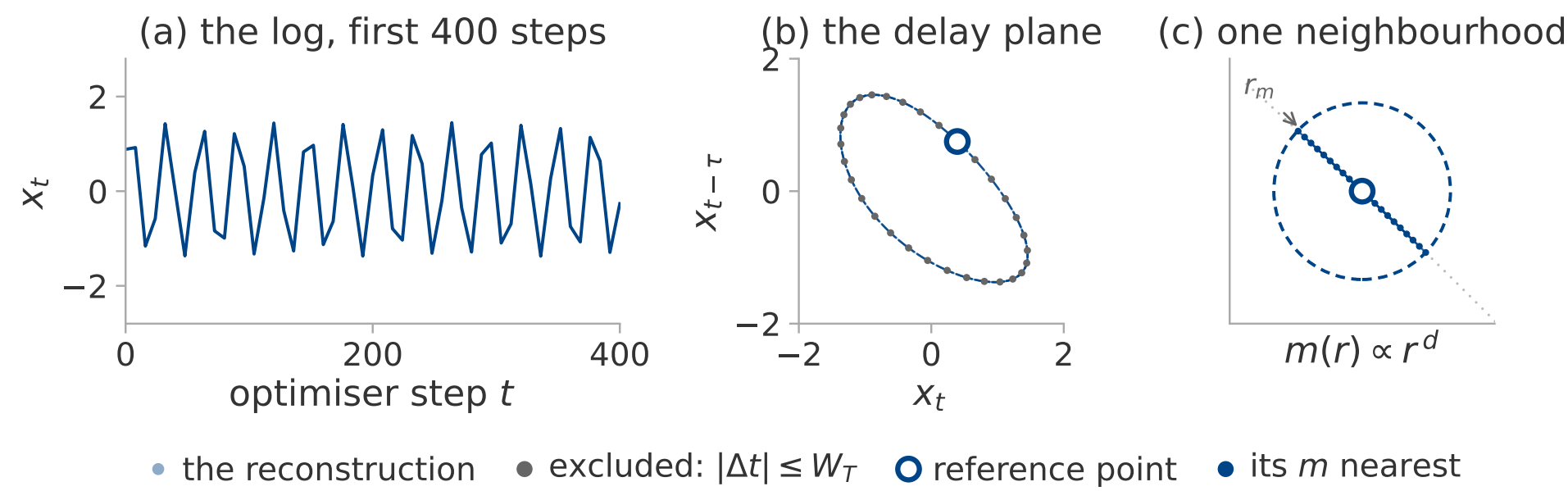
Let $\theta = \theta_0 + V^\top c$, where $V \in \mathbb{R}^{k \times P}$, $VV^\top = I_k$ and $c \in \mathbb{R}^k$. Let $f(c)$ be the outputs on a fixed probe set.

Let μ_R be the occupation measure induced by an ergodic invariant measure of regime R , assumed exactly dimensional. Write $C_W = \text{Cov}_{t \in \mathcal{W}}(c_t)$ for a trajectory window $\mathcal{W}$.

Quantity	Definition
Available dimension	$d_{\text{avail}} = \dim \text{im}(V^\top) = k$.
Functional dimension	$d_{\text{func}}(c) = \text{rank}\left(\frac{\partial f(c)}{\partial c}\right)$.
Active dimension	$d_{\text{act}}(R) = \lim_{\varepsilon \downarrow 0} \frac{\log \mu_R(B(c, \varepsilon))}{\log \varepsilon}$, μ_R -a.e.
Covariance effective rank	$\text{PR}^{\text{pos}} = \frac{(\text{tr } C_W)^2}{\text{tr}(C_W^2)}$, $C_W \neq 0$.

$B(c, \varepsilon)$ is a Euclidean ball. For PR^{det} , replace c_t by its residual about the window's least-squares line. Finite samples resolve finite-scale exponents, not the limit defining d_{act} .

02 Delay embedding and dimension estimation



One scalar record, its delay reconstruction, and a neighbourhood. The loop is visited repeatedly.

$$y_t = (x_t, x_{t-\tau}, \dots, x_{t-(E-1)\tau}) \in \mathbb{R}^E,$$

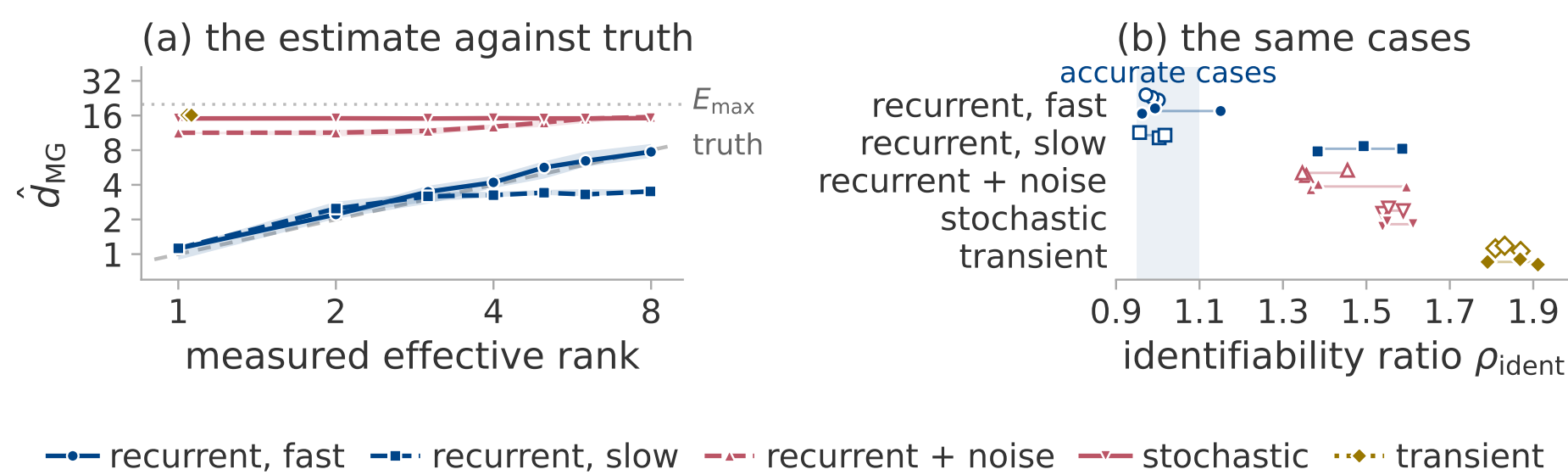
$$\hat{d}_{\text{MG}}^{-1} = \frac{1}{N(m-1)} \sum_{i=1}^N \sum_{j=1}^{m-1} \log \frac{r_m^{(i)}}{r_j^{(i)}}.$$

Pool the nearest-neighbour likelihood over N delay vectors, each with m neighbours, before inversion. $r_j^{(i)}$ is the j th neighbour distance; the **Theiler exclusion** requires time separation greater than W_T .

A dimensional interpretation needs a valid delay embedding, one regime per window, enough returns after exclusion, and a stable scaling range.

03 Recovery accuracy across dynamical regimes

A linear head on a frozen image network is forced by r independent quasiperiodic phases. This supplies known active dimension. Settings are frozen before testing held-out ranks and seeds.



Left: recovery by regime; markers are medians over seeds and observers, bands are interquartile ranges. Right: stability under doubled embedding, at $r = 2$ (open) and $r = 6$ (filled).

- Recurrent:** about one-component error up to roughly eight components at the main configuration.
- Stochastic:** noise shifts the value and makes it depend on embedding settings.
- Transient:** temporal exclusion can produce a large spurious value.

Only one learned-system benchmark passes every prescribed check. All validation systems are externally forced.

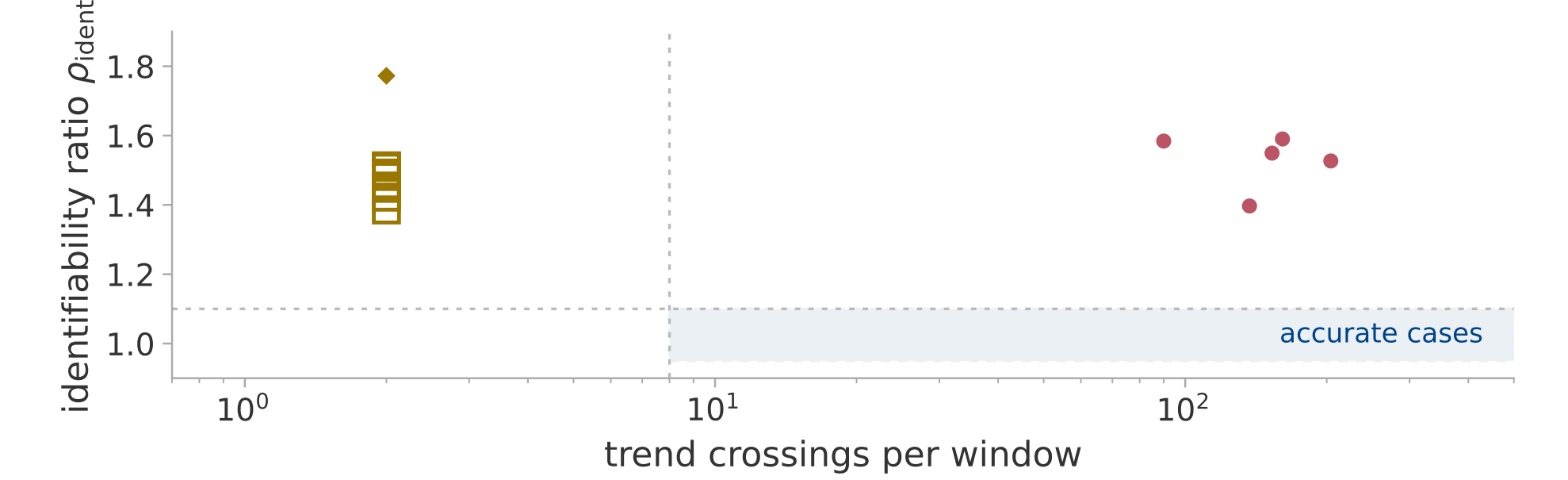
06 Estimation limits and experimental confounders

- Recovery depends on the lag, observer and window. The tested resolution ceiling is low.
- The grokking comparison has four regularised runs against two without weight decay. Regularisation and outcome are confounded.

Interpret the geometry only after checking the regime. For these grokking runs, the evidence supports a reversible redistribution of trajectory variance.

04 Scalar-log diagnostics in grokking

Grokking is delayed generalisation after memorisation. We examine a mini-batch transformer and an unregularised full-batch perceptron.



Each point is a log. The blue band records accurate controlled cases; it is not a validated decision boundary.

The **identifiability ratio**

$$\rho_{\text{ident}} = \frac{\hat{d}_{\text{MG}}(2E_{\text{max}})}{\hat{d}_{\text{MG}}(E_{\text{max}})}$$

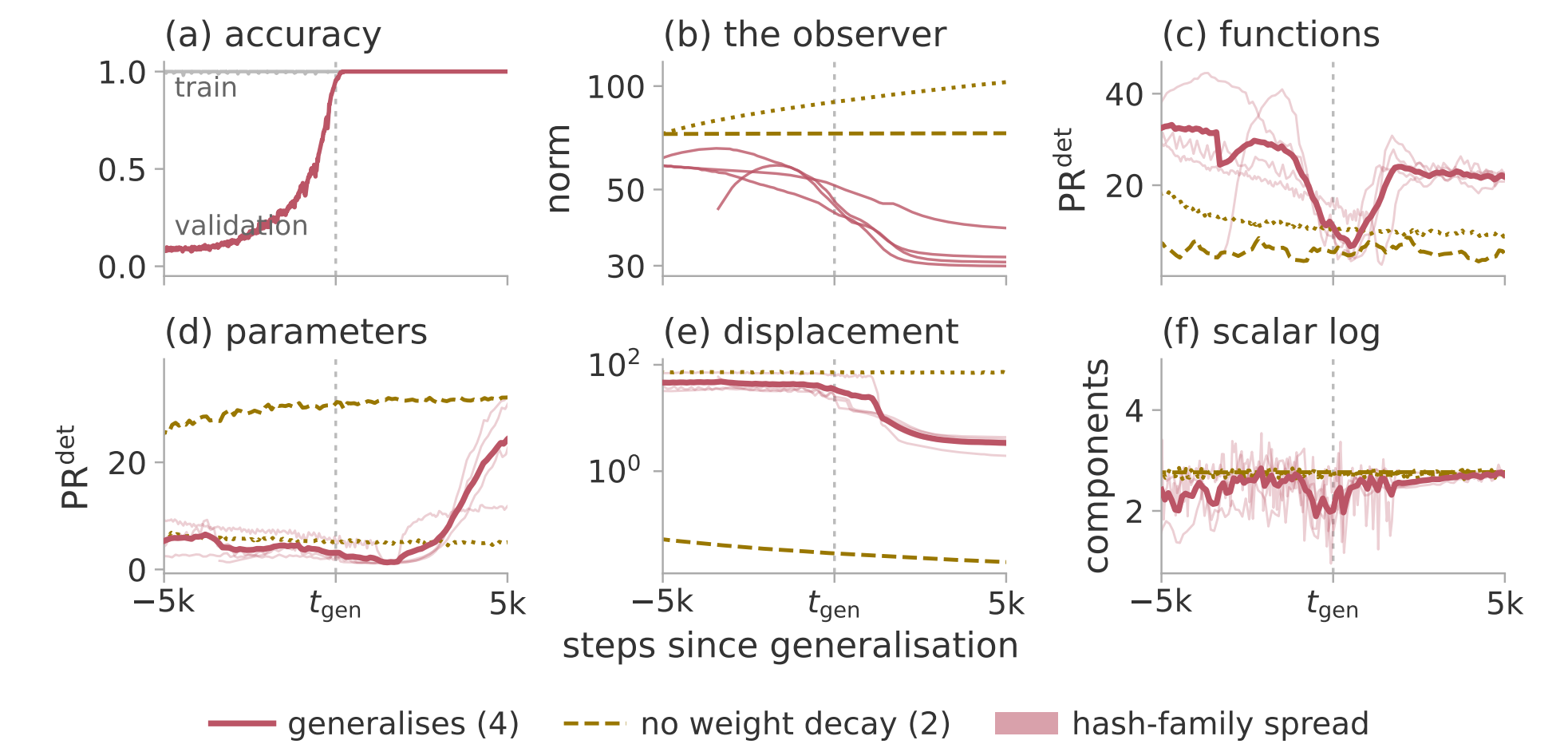
tests sensitivity to embedding dimension. Trend crossings measure non-monotonicity of the log.

All evaluated grokking logs lie outside the reference region. Their scalar levels are not interpreted as active dimension. These diagnostics test neither recurrence itself nor the scaling range.

05 Covariance effective rank near generalisation

We record parameter and probe-logit trajectories through 1024-dimensional CountSketch projections. In each window, detrended covariance eigenvalues λ_i give

$$\text{PR}^{\text{det}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$



Four regularised runs (rose): three modular-addition seeds and one S_5 task. Two runs without weight decay (gold) are aligned to paired transition times. Thick rose curves average four runs; shading is hash-family spread, not uncertainty across runs.

- Covariance effective rank falls near generalisation and later re-expands in parameter and function space.
- A window matched to the transition gives a correlated drop in the parameter-norm log estimate.

Effective rank can fall while active dimension stays fixed. The scalar drop describes a transition retrospectively; it does not establish early prediction.

